

QUOKKA-based understanding of outflows (QED) - IV. Limitations of $H\alpha$ as an outflow diagnostic

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Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

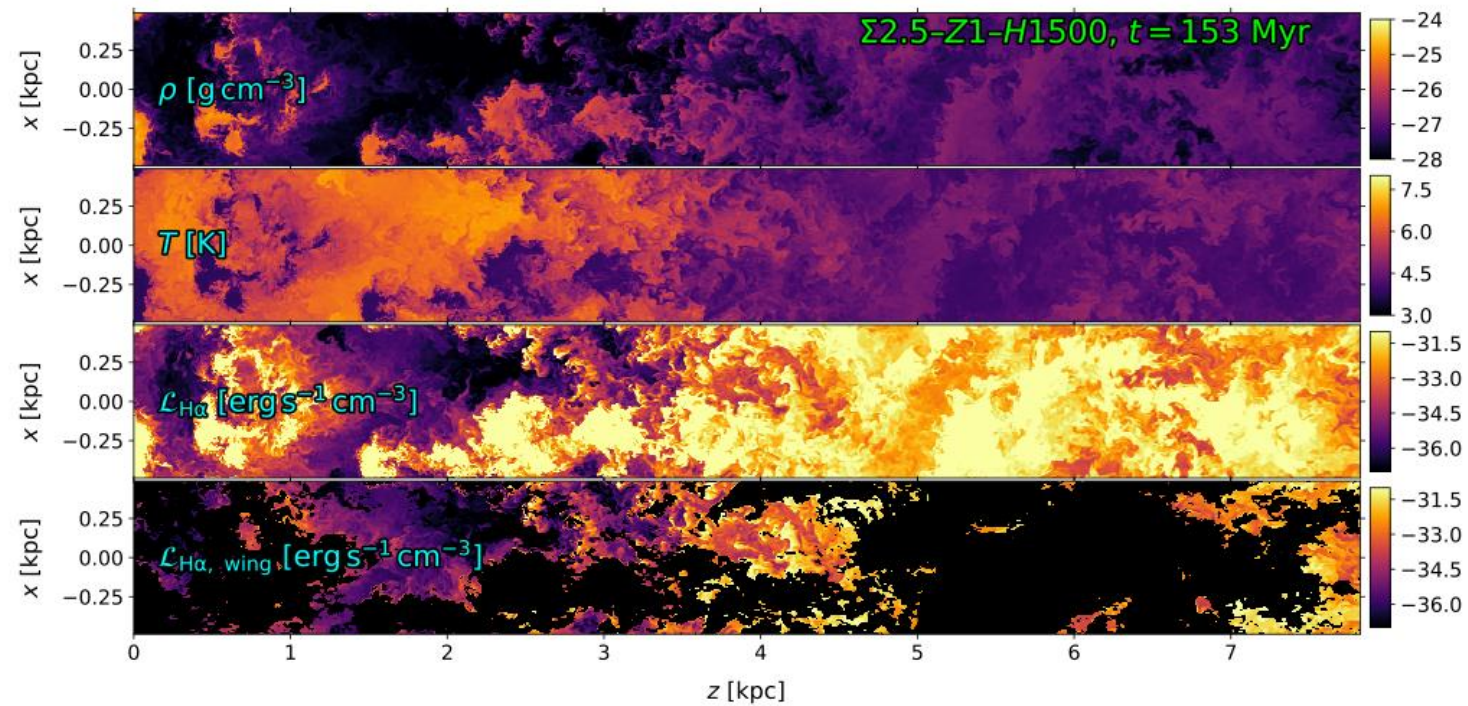
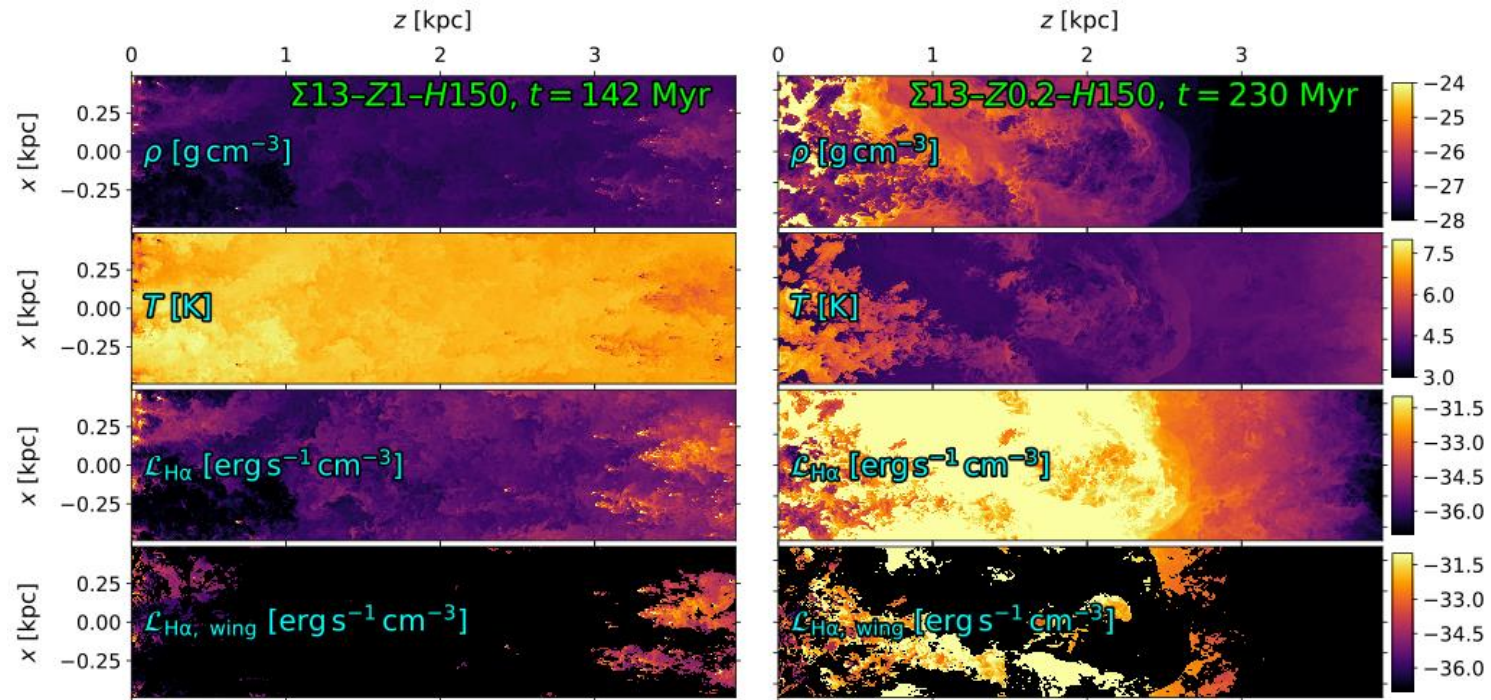
The presence of broad wings in the $H\alpha$ line is commonly used as a diagnostic of the presence and properties of galactic winds from star-forming galaxies. However, the accuracy of this approach has not been subjected to extensive testing. In this paper, we use high-resolution simulations of galactic wind launching to calibrate the extent to which broad $H\alpha$ wings can be used to infer the properties of galactic outflows. For this purpose, we analyse a series of high-resolution wind simulations from the QED suite spanning two orders of magnitude in star formation surface density (Σ_{SFR}). We show that the broad component of $H\alpha$ emission correlates well with the wind mass flux at heights ~ 1 kpc above the galactic plane, but that the correlation is poor at larger distances from the plane, and that even at 1 kpc the relationship between mass flux and surface brightness of broad $H\alpha$ is significantly sub-linear. The sub-linear scaling suggests that the electron column density in the wind increases systematically with outflow strength, and that the conventional assumption of constant electron density in the wind leads to a systematic overestimate of how steeply mass loading factors depend on Σ_{SFR} . We provide empirical scaling relations that observers can apply to correct for this effect when converting $H\alpha$ measurements to mass outflow rates. Finally, we use synthetic observations of the density-diagnostic [S II] $\lambda\lambda 6716, 6731$ doublet to show that using this diagnostic only slightly improves estimates of wind outflow rates compared to the naive assumption of constant electron density, and performs significantly worse than the empirical correlation we provide.

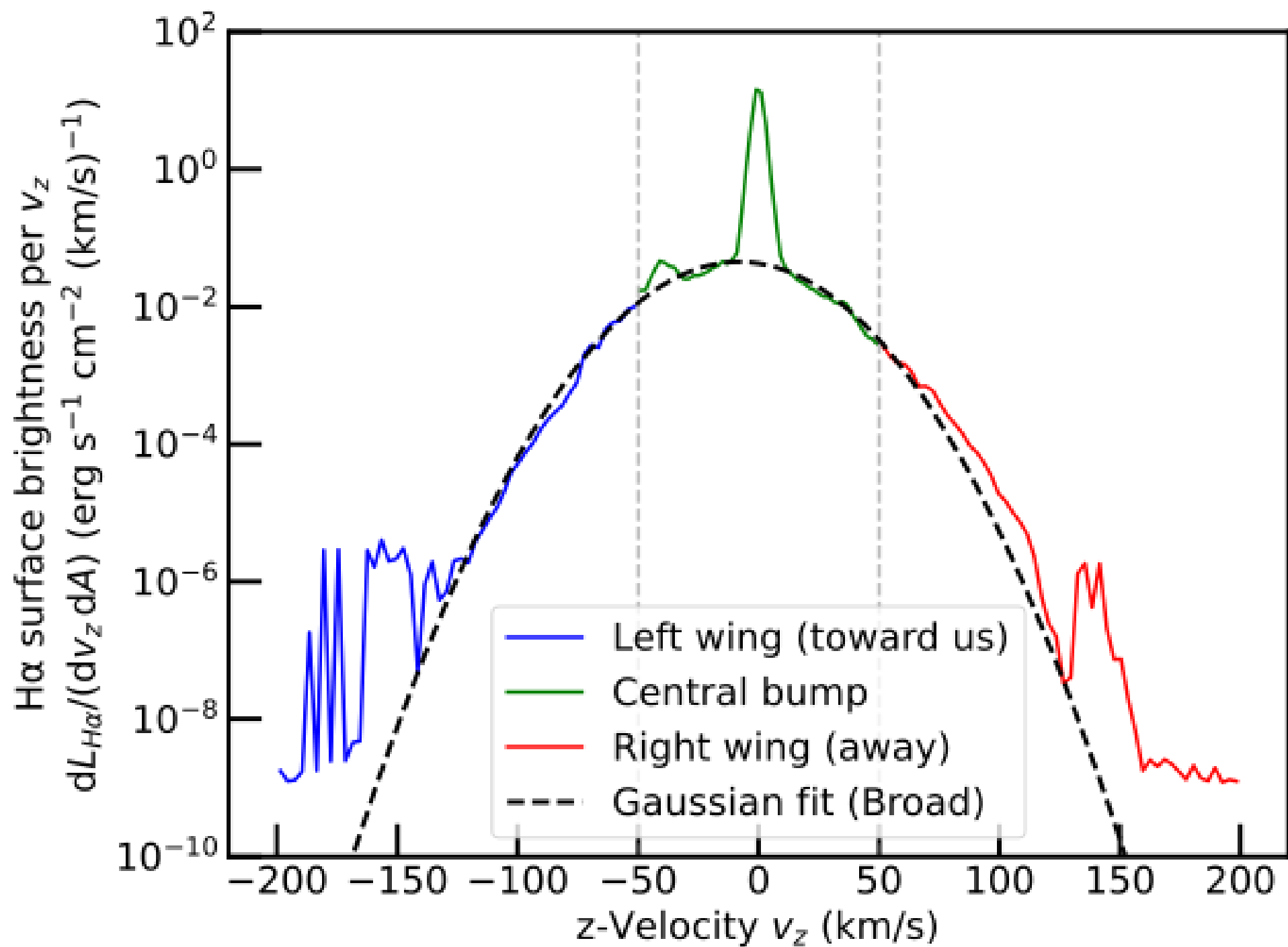
Key words: ISM: jets and outflows — ISM: structure — galaxies: ISM — line: profiles — methods: data analysis

Name	Σ_{SFR} [$M_{\odot} \text{ yr}^{-1} \text{ kpc}^{-2}$]	Δx [pc]	L_z [kpc]	σ_{broad} [km s^{-1}]	$\Sigma_{H\alpha, \text{broad}}$ [$10^{37} \text{ erg s}^{-1} \text{ kpc}^{-2}$]	t [Myr]
(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\Sigma 13\text{-Z1-H150}$	6×10^{-3}	2	4	25.42	9.18	142
				31.48	7.15	160
				42.13	7.88	181
				56.48	8.31	210
				43.08	12.0	221
$\Sigma 13\text{-Z0.2-H150}$	6×10^{-3}	2	4	20.97	15.2	230
				14.13	1.70	250
				25.32	3.40	270
				23.76	3.14	290
$\Sigma 2.5\text{-Z1-H1500}$	1.58×10^{-4}	4	8	18.03	0.0384	153
				19.93	0.0691	200
				20.96	0.0407	249
				23.38	0.0503	299

Table 1. Simulation parameters for the three QED runs used in this work (including a subset of Table 1 in QED III). Columns give: (1) run name; (2) star-formation surface density Σ_{SFR} ; (3) cell size Δx ; (4) vertical box half-height L_z ; (5) dispersion of the broad $H\alpha$ emission from the wing-only Gaussian fit, σ_{broad} ; (6) broad-component $H\alpha$ surface density $\Sigma_{H\alpha, \text{broad}}$ (see Section 2.3); and (7) snapshot time t . Units are indicated in the header.

¹ QUOKKA-based Understanding of Outflows Derived from Extensive, Repeated, Accurate, Thorough, Demanding, Expensive, Memory-consuming, Ongoing Numerical Simulations of Transport, Removal, Accretion, Nucleosynthesis, Deposition, and Uplifting of Metals (QUOD ERAT DEMONSTRANDUM, or QED).





arrive at a relationship between the broad wing luminosity and the outflow rate in this simple slab model:

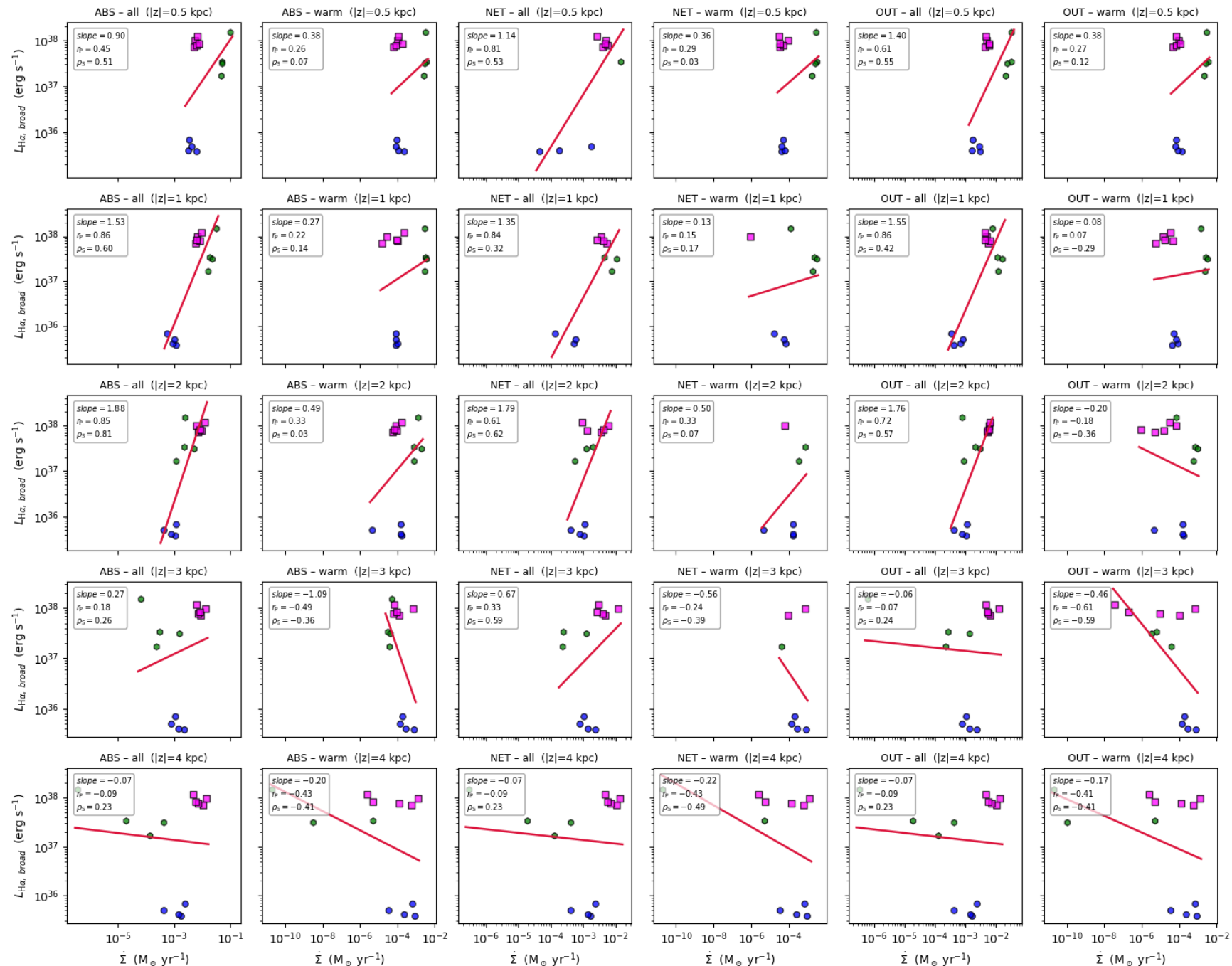
$$\dot{\Sigma}_{\text{out}, \text{H}\alpha} = \frac{\mu m_{\text{H}} \sigma_{\text{broad}}}{\alpha_{\text{H}\alpha, \text{eff}} h \nu_{\text{H}\alpha} N_e} \Sigma_{\text{H}\alpha, \text{broad}} \quad (7)$$

where $N_e = n_e H$ is the column density through one side of the slab.

turn in Section 3.3. A fit for the relation between $\dot{\Sigma}_{\text{out}}$ and $\Sigma_{\text{H}\alpha, \text{broad}}$ at $|z| = 1$ kpc, where the correlation is strongest, gives

$$\dot{\Sigma}_{\text{out}} = 0.022 \times \left(\frac{\Sigma_{\text{H}\alpha, \text{broad}}}{10^{40} \text{ erg s}^{-1} \text{ kpc}^{-2}} \right)^{0.47} M_{\odot} \text{ yr}^{-1} \text{ kpc}^{-2}, \quad (11)$$

with a Pearson coefficient $r_p = 0.86$, and at other heights ≤ 2 kpc the slope of the correlation lies in the range $\approx 0.3 - 0.5$.



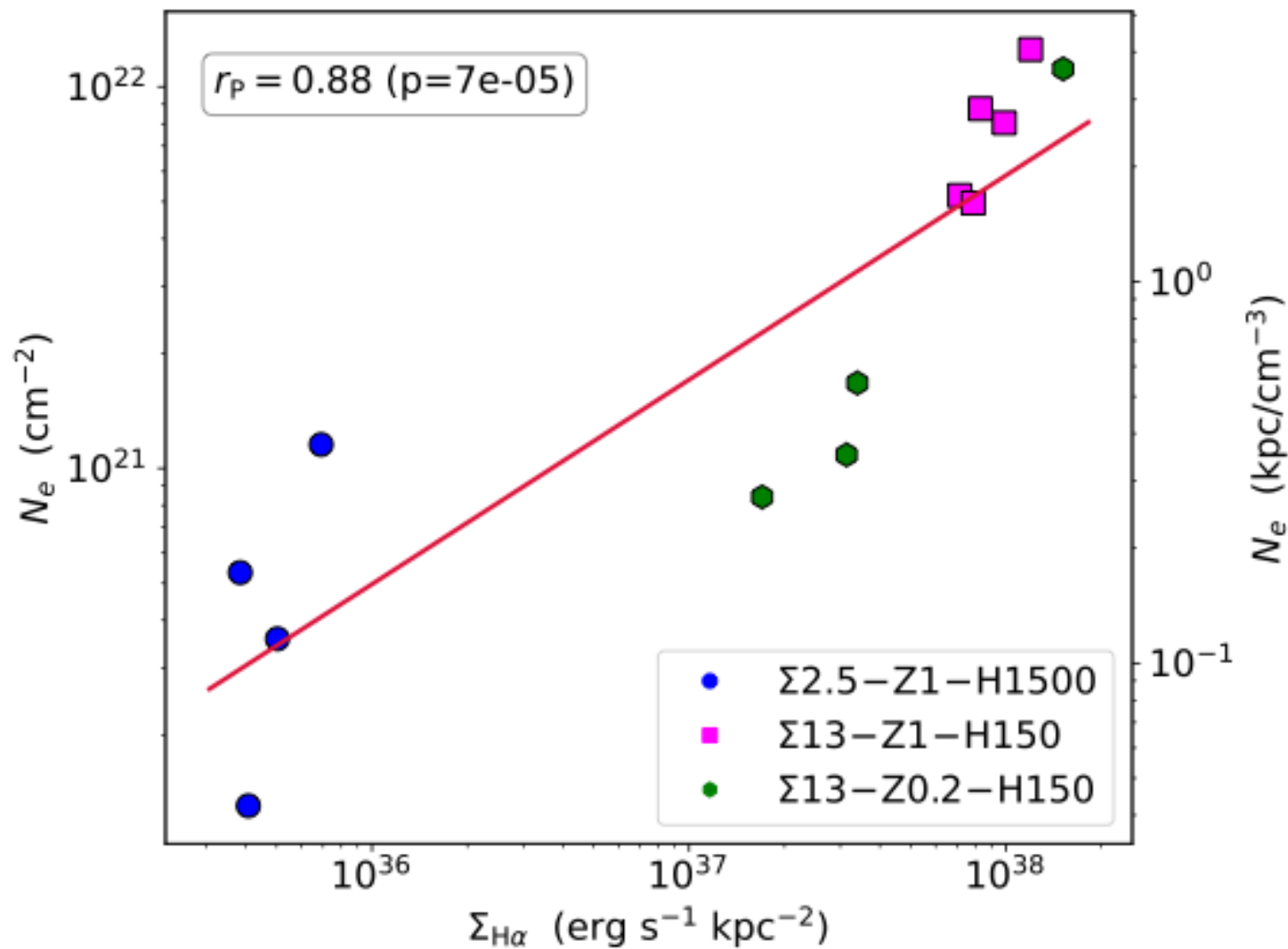


Figure 6 show the value of N_e required to obtain the correct out-flow rate at $|z| = 1$ kpc as a function of the broad component surface brightness $\Sigma_{\text{H}\alpha, \text{broad}}$. However, the results are qualitatively similar for any height $\lesssim 2$ kpc. The relationship exhibits a tight correlation with a Pearson coefficient $r_P = 0.88$, and an OLS fit to the data gives a relation

$$N_e = N_{e,40} \left(\frac{\Sigma_{\text{H}\alpha, \text{broad}}}{10^{40} \text{ erg s}^{-1} \text{kpc}^{-2}} \right)^{0.54} \quad (12)$$

